

Restaurant Spending Analysis

Plotly Express tips dataset | Python | Customer Analytics

Objective

Examine how restaurant spending varies by day, smoking status and meal period, and assess whether party size is associated with total bill.

Dataset and methods

The dataset contains **244 restaurant bills** with total_bill, tip, sex, smoker, day, time and size. Methods include descriptive statistics, Welch independent-samples t-test, Chi-square testing, simple linear regression and residual diagnostics.

Business interpretation

Party size is relevant to table allocation, expected spend and capacity planning. Group differences should be interpreted as associations, not causal effects.

Reproducibility

The project is designed as a compact Python analytics portfolio example covering cleaning, descriptive analysis, hypothesis testing, regression and model diagnostics. The repository includes the analysis script, requirements and supporting figures.